

Astro — Telegram Quick Reference

Sony a6400 · pi4-astro-oz · send `/astro` to your bot for this list

CAPTURE

<code>/astro capture</code>	Runs the active profile
<code>/astro capture profile=<name></code>	Use a named profile
<code>/astro capture exposure=S iso=N frames=N</code>	Override anything
<code>/astro capture interval=S duration=S</code>	...makes it a timelapse
<code>/astro preview exposure=S iso=N</code>	Quick daylight frame
<code>/astro last</code>	Resend last frame
<code>/astro abort</code>	Stop the run
<code>/astro status</code>	State, camera, active profile

Abort waits. It stops between frames — an exposure already running finishes first.

PROFILES

<code>/astro profiles</code>	List them all
<code>/astro profile</code>	Show active settings
<code>/astro profile exposure=S iso=N</code>	Edit active settings
<code>/astro profile load <name></code>	Make it active
<code>/astro profile save <name></code>	Save active as a profile
<code>/astro profile add <name> k=v...</code>	Define one
<code>/astro profile rm <name></code>	Delete it

BUILT-IN PROFILES

SINGLE SHOT	EXPOSURE	ISO
daylight	1/500	100
overcast	1/250	200
indoor	1/60	1600
focus-check	1/60	6400
moon	1/125	100
milky-way	20 s	3200
deep-sky	30 s ×20	1600

TIMELAPSE	RUN	EVERY	ISO	EXP
timelapse-sunset	90 min	6 s	100→3200	0.5→4 s
timelapse-sunrise	90 min	6 s	3200→100	4→0.5 s
timelapse-blue-hour	40 min	6 s	400→1600	1→3 s
timelapse-night-sky	2 hr	30 s	3200	15 s
timelapse-stars	3 hr	30 s	6400	20 s
timelapse-golden-hour	60 min	5 s	100	0.01 s

The `timelapse-` prefix is only a label. What makes a run a sequence is the interval it carries, plus a bound — frames or duration. Partial timing is rejected, not guessed.

EXPOSURE RAMPING

<code>iso_start= iso_end=</code>	ISO across the ramp
<code>exposure_start= exposure_end=</code>	Exposure across the ramp
<code>ramp_duration=S</code>	Ramp, then hold at the end values

Works on any profile or command. Ramps logarithmically, so 100→200 takes the same frames as 1600→3200.

FROM CAPTURE TO VIDEO

`capture` → `archive` `session` → `render`

Frames land on the Pi. **Archiving is manual** — and the Pi keeps only the newest 200 frames, so archive a long run before it prunes itself.

<code>/astro sessions</code>	Runs on the Pi, by id
<code>/astro shots <id></code>	Frames + thumbnails
<code>/astro archive session <id></code>	Copy a run to the LXD
<code>/astro archive last <frame></code>	Copy one frame
<code>/astro archived [id]</code>	Browse the archive
<code>/astro render <id></code>	Stitch a run into MP4
<code>/astro delete session <id> confirm</code>	Delete from the Pi
<code>/astro delete all confirm</code>	Delete everything

CAMERA

<code>/astro camera</code>	Camera, limits, defaults
<code>/astro camera a6400 a6700</code>	Switch body
<code>/astro defaults k=v</code>	Per-camera baseline
<code>/astro defaults load</code>	Reset active to baseline
<code>/astro battery</code>	Battery level
<code>/astro limits</code>	Exposure and ISO limits — outside them is an error, never a silent clamp
<code>/astro exposures</code>	Valid shutter speeds
<code>/astro isos</code>	Valid ISO values

BEFORE A SESSION

1. Camera on, dial to **M**.
2. Menu → Setup → USB Connection → **PC Remote**. If it enumerates as “charging mode”, gphoto2 cannot see it.
3. `/astro battery` to confirm the link.

RULES OF THUMB

500 rule. Star trails start at $\approx 500 \div (\text{focal} \times 1.5)$ on this APS-C body — the crop factor is easy to forget. 20 mm → ~17 s, not 25.
NPF is stricter. For pin-point stars at 100%, 20 mm f/2.8 → ~11 s. The 500 rule was written for film.
Bulb is automatic above 30 s.
Mount control is not implemented — `/astro track` says so.